开源软件基础

基于 z3 的约束求解

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Outline

z3 简介

概念

Z3 is a theorem prover from Microsoft Research. It is licensed under the MIT license[a].

[a] https://github.com/Z3Prover/z3

约束求解的历史

1. SAT: The satisfiability problem
2. SMT: Satisfiability modulo theories

可满足问题

SAT

可满足性（Satisfiability）是用来解决给定的真值方程式，是否存在一组变量赋值，使问题为可满足。布尔可满足性问题（Boolean satisfiability problem；SAT)）属于决定性问题，也是第一个被证明属于 NP 完全的问题。此问题在计算机科学上许多领域的皆相当重要，包括计算机科学基础理论、算法、人工智能、硬件设计等等。[a]

[a] https://zh.wikipedia.org/zh-cn/

Outline

约束求解示例

今有雉兔同笼，上有三十五头，下有九十四足，问雉兔各几何？

约束求解示例

上置头，下置足，半其足，以头除足，以足除头，即得。

```
num_heads = 35
num_feet = 94
num_feet_half = num_feet // 2
num_rabbits = num_feet_half - num_heads
num_chicken = num_heads - num_rabbits
```

孙子算经

约束求解示例

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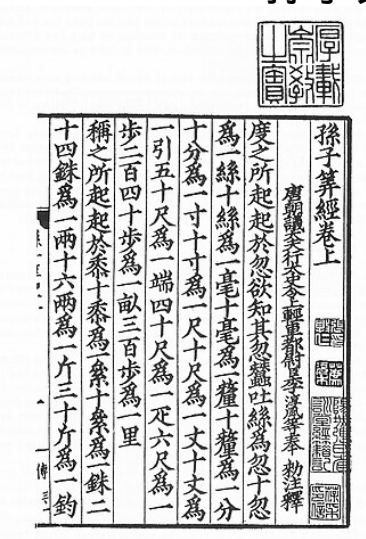
孙子算经


孫子算經卷上

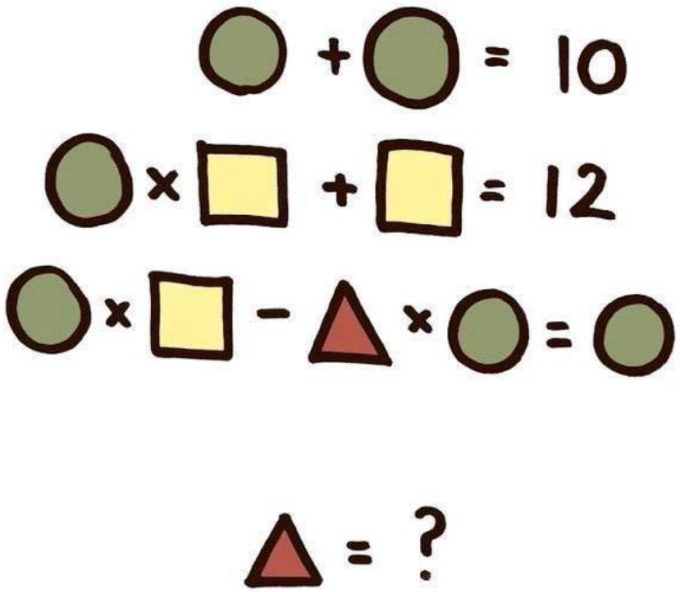
度之所起起於忽欲知其忽纖吐絲爲忽十忽爲一絲十絲爲一毫十毫爲一釐十釐爲一分十分爲一寸十寸爲一尺十尺爲一丈十丈爲一引五十尺爲一端四十尺爲一疋六尺爲一步二百四十步爲一畝三百步爲一里

稱之所起起於黍十黍爲一絫十絫爲一銖二十四銖爲一兩十六兩爲一斤三十斤爲一鈞

约束求解示例

```
import z3
chicken, rabbits = z3.Ints('chicken rabbits')
z3.solve(chicken >= 1,    # number of chicken
      rabbits >= 1,       # number of rabbits
      chicken + rabbits == 35,
      chicken * 2 + rabbits * 4 == 94)
```

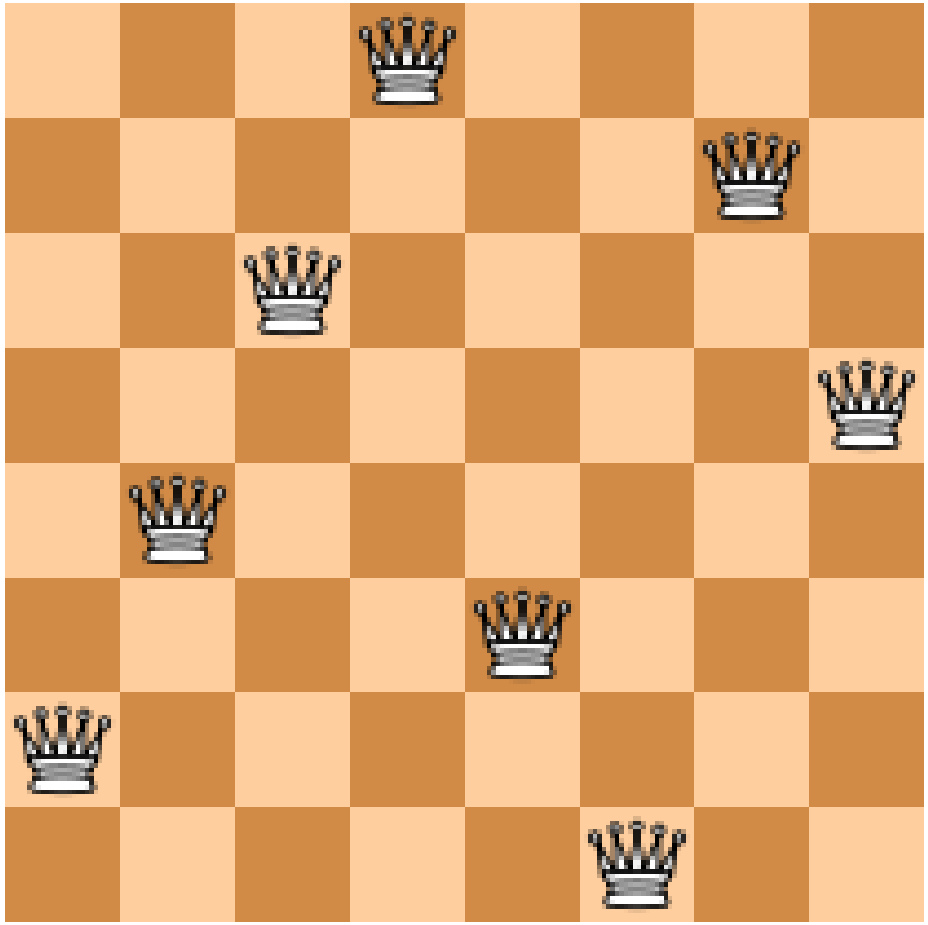
约束求解示例



约束求解示例

```
from z3 import *
circle, square, triangle = Ints('circle square triangle')
s = Solver()
s.add(circle+circle==10)
s.add(circle*square+square==12)
s.add(circle*square-triangle*circle==circle)
print(s.check())
print(s.model())
```

八皇后问题



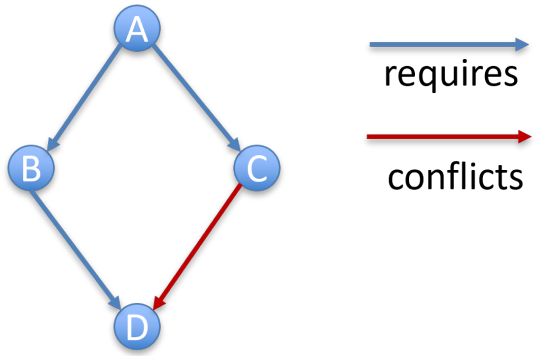
八皇后问题

```
Q = [ Int('Q_%i' % (i + 1)) for i in range(8) ]

# Each queen is in a column {1, ... 8 }
val_c = [ And(1 <= Q[i], Q[i] <= 8) for i in range(8) ]

# At most one queen per column
col_c = [ Distinct(Q) ]

# Diagonal constraint
diag_c = [ If(i == j,
              True,
              And(Q[i] - Q[j] != i - j, Q[i] - Q[j] != j - i))
           for i in range(8) for j in range(i) ]

solve(val_c + col_c + diag_c)
```

程序生成/修复

```
from z3 import *
x = Int('x')
y = Int('y')
f = Function('f', IntSort(), IntSort())
s = Solver()
s.add(f(f(x)) == x, f(x) == y, x != y)
print(s.check())
m = s.model()
print("f(f(x)) =", m.evaluate(f(f(x))))
print("f(x)    =", m.evaluate(f(x)))
```

Encoding Dependencies to Z3




```
import z3
A, B, C, D = z3.Bools("A B C D")
p1, p2, p3, p4, p5, p6 = z3.Bools("p1 p2 p3 p4 p5 p6")
solver = z3.Solver()
solver.assert_and_track(-1 * z3.If(A, 1, 0) + z3.If(B, 1, 0) >= 0, p1)
solver.assert_and_track(-1 * z3.If(A, 1, 0) + z3.If(C, 1, 0) >= 0, p2)
solver.assert_and_track(-1 * z3.If(A, 1, 0) + z3.If(D, 1, 0) >= 0, p3)
solver.assert_and_track(-1 * z3.If(B, 1, 0) + z3.If(D, 1, 0) >= 0, p4)
solver.assert_and_track(z3.If(C, 1, 0) + z3.If(D, 1, 0) <= 1, p5)
solver.assert_and_track(A == True, p6)

print(solver.check())
print(solver.unsat_core())
```